

ILLINOIS TECH | College of Computing

Ch10: The Digital Divide, Democracy, And Work P2: Disability, Race, & Gender



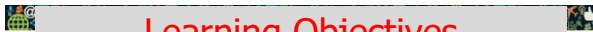
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Announcements

- Internet 02 section- clarification
- Final Exam

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Learning Objectives

Upon completion of this lesson the students should be able to:

- Define the **Digital Divide**
 - ❖ Describe and discuss the moral and ethical issues that define and surround the Digital Divide and proposals for its resolution
- Describe and discuss issues surrounding **making technology available to people with disabilities**
- Discuss the effect of **technology on racism**
- Discuss the impact of **technology on gender issues**
 - ❖ Describe and discuss gender issues in employment in the technology field
- Explain and discuss the impact of **technology on democracy and democratic values**
- Describe how technology has **transformed the nature of work and discuss its impact on the quality of work life**

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Module Objectives

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Cybertechnology and the Disabled

- The World Wide Web Consortium (W3C) was formed to promote standards that ensure universal Web access.
- It established a Web Accessibility Initiative (WAI), which has produced guidelines and protocols for developing software applications that improve access for disabled persons.

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Cybertechnology and the Disabled

- WAI applications range from software used in speech synthesizers and screen magnifiers to proposed applications that will benefit disabilities that are:
 - visual,
 - hearing,
 - physical,
 - cognitive,
 - neurological.

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Cybertechnology and the Disabled

- WAI has established guidelines for developing “user agents” that are:
- intended to lower barriers to Web access for people with disabilities.
- designed to conform and communicate with “assistive technologies” such as screen readers (performing a function similar to Braille in off-line contexts).

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Cybertechnology and the Disabled

- Grodzinsky (2000) **argues that** computers equipped with assistive technologies and “adaptive devices” can be an “equalizer” in the era of information technology.
- For example, they enable disabled persons to participate in and compete for jobs that require computer access.

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Arguments for Continued WAI Support for Disabled Persons

- Despite questions surrounding the potential downside of remote-work practices for disabled persons, there are many examples in which WAI’s assistive and adaptive technologies have clearly benefited disabled workers.
- So, on balance, WAI has made a difference for many disabled persons who desire to compete for jobs in the contemporary workplace.
- But some critics might ask why we should continue to fund WAI initiatives, which can be very expensive, especially given that disabled persons comprise a relatively small portion of the overall population.

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Arguments for Supporting WAI

- WAI proponent Cheiko Asakawa (2012) argues that access to technology is not simply a privilege but rather a “human right.”
- Other WAI supporters point out that some technology-related measures taken for the disabled have had positive outcomes for other groups.
- Some also believe that voice recognition technology designed to assist disabled persons who are unable to use keyboards will ultimately benefit nondisabled persons with low literacy skills.

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Arguments for Supporting WAI

- WAI’s advocates also point out that measures taken for the disabled have had positive outcomes for other groups – for example:
- poor people who often are forced to deal with literacy problems have benefited;
- ramps designed for wheelchair accessibility not only benefit people in wheelchairs, but also non-disabled persons as well (such as parents pushing baby carriages).

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Arguments for Supporting WAI

- Some argue that ordinary users will likely continue to benefit from design enhancements to user interfaces intended to assist disabled persons.
- But there is a danger in using this kind of utilitarian argument as the main rationale.
- For example, suppose that some enhancements do not benefit non-disabled persons as well?
- Should we then cease to fund research for those kinds of technologies that support the disabled?

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Race and Cybertechnology

- Background: Consider some statistics ranging from 2000 to 2011 that correlate race with the digital divide in the United States.
- In 2000, 51% of all homes had at least one computer, and 41.5% of all homes had Internet access.
- From the vantage point of race, 46.1% of white Americans and 56.8% of Asian Americans and Pacific Islanders had access, contrasted with only 23.5% of African Americans and 23.1% of Hispanics who did (in 2000).

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Race and Cybertechnology

- By 2011, statistics for African American vs. white users had changed significantly.
- Whereas Internet usage among whites was estimated to be 88%, the rate of African Americans using the Internet had grown to 80%.
- Perhaps even more interesting was the rate at which the use of access to broadband Internet connection had grown for African Americans.
- Whereas 65% of African American Internet users had broadband access, only 53% of white American Internet users enjoyed this service.

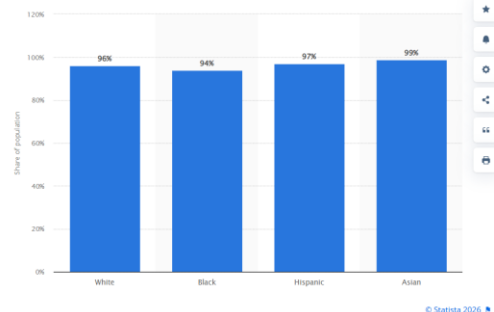
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Race and Cybertechnology

- We focus mainly on two different kinds of issues affecting race and cybertechnology:
1. Internet usage patterns among minority groups;
 2. Some ways in which the Internet has been used as a medium to spread racial prejudice.

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Share of adults in the United States who use the internet as of June 2025, by ethnicity



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Race and Cybertechnology Issues Affecting Internet Usage Patterns

- Kretchmer and Karveth (2001) note that in studies involving African-Americans:
- the average age for African-American Internet users tends to be younger than for whites;
- African Americans typically tend to access the Internet less frequently than whites;
- adult African American Internet users are much more likely than their white counterparts to have modest incomes, no college degrees, and children under eighteen.

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Race and Cybertechnology Issues Affecting Usage Patterns

- Kretchmer and Karveth also claim that African-American Internet users are:
- *more likely* to use the Internet for entertainment and for locating information about quality-of-life activities, such as job training, school, health care, and hobbies;
- *less likely* to participate in Web-based auctions, and to use e-mail to develop and sustain friendships.

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The Internet as a Medium to Spread Racist Speech

- Theismeyer (1999) believes that two distinct kinds of racist speech can be found on the Internet:
- (a) *hate speech* itself, which can include text, music, online broadcast, and images that exhort users to act against targeted groups;
- (b) *persuasive rhetoric* that does not directly enunciate racism and corresponding violence, but which ultimately promotes or justifies it.

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Racism and Rhetoric on the Internet

- Theismeyer concludes that it is impossible to know at this point whether cyber technology has been the main cause of the rapid spread of racism, especially the neo-Nazism movement, in recent years.
- However, she is convinced that cyber-technology has been its principal tool.

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Gender and Cybertechnology

- Pew Internet-Project surveys conducted in 2005 also showed that in the United States, 66% of women went online, as opposed to 68% of men.
- But one possible reason why the percentage of male Internet users in the U.S. was higher than female users (in 2005) is because women make up a greater share of the overall U.S. population.
- By 2008, the proportion of women (overall) who used the Internet was equal to that of men.

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Racism and Rhetoric on the Internet

- In her analysis of racist rhetoric on the Internet, Theismyer examines two questions:
 - I. Does cyber technology (or what she calls “information technology”) make the reemergence of prejudicial messages and attitudes swifter and more likely?
 - II. Does the Internet's wide range of distribution make for more followers of racial prejudice, and finally more persuasion?

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Gender and Cybertechnology

- The gap that has traditionally existed between the percentage of female and male Internet users in the United States has changed significantly since some studies that were conducted in (or before) 2000.
- According to a 2005 report by the *Pew Internet & American Life Project*, young women were slightly more likely to be online than young men, and the number of black women online surged between 2002 and 2005 (black women who used the Internet outnumbered black men by about 10%).

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Gender and Cybertechnology

- While the gap between female and male Internet users has virtually disappeared in the U.S., and has narrowed considerably in many other Western countries as well, this has not been the case globally.
- Shade (2002) describes how some women in the Philippines, Latin America, Africa, and Asia have developed “grass-roots” initiatives, which she refers to as “globalizing from below” to address the technology gap.
- She also notes that at the second Global Knowledge Conference, held in Malaysia in 2000, specific initiatives were introduced to support gender equity and women's empowerment using cyber technology.

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Gender and Cybertechnology

- Consider two distinct kinds of issues affecting gender:
 1. access issues (i.e., access to jobs for women in the computing/engineering fields);
 2. gender bias in software applications, including video-games.

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Access Issues for Women

- Access-related issues affecting computing include the “pipeline” for women entering the field of computer science (CS).
- In a classic study, Camp (1997) noted that the number of women getting PhDs in CS had increased slightly.
- However, Camp also noted that the number of women getting BS degrees in CS had decreased.

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Gender and Access to High-Tech Jobs

- Kirlidog, Aykol, and Gulscecen (2009) cite more recent evidence to support the ongoing concerns about the “pipeline.”
- They argue that computer science is still typically regarded as a “male profession,” both in industry and academia.
- They also believe that women remain in the “margins” of a male-dominated profession, which is filled with highly gendered expressions such as “killing or aborting programs,” “workbench,” “toolkit,” etc., that reflect the masculine culture of the field.

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Gender and Access to High-Tech Jobs

- Kirlidog, et al. also identify three “net results” of the “male-dominated computing profession,” in which they claim women are:
 - 1) under-represented in computer-related jobs;
 - 2) more under-represented in the managerial ranks in the computing field because of the ‘glass ceiling’;
 - 3) earn less than men for doing the same jobs.

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Gender Bias in Video Games

- Does gender bias exist in software development for video games technology?
- Buchanan (2000) argues that video games contribute to gender bias because many of them tend to:
 - 1) *misrepresent* or to *exclude* female characters;
 - 2) perpetuate traditional *sexist stereotypes*.

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Gender Bias in Video Games

- Initially, we might be inclined to dismiss concerns about gender bias in video games on the grounds that many women aren’t interested in these applications.
- But Brey (2008) argues that the question of gender bias in video games is “morally significant” because women will have less opportunity to enjoy computer games and their possible benefits, if computer games tend to be designed and marketed for men.
- The “benefits,” for which women would have less opportunity include greater “computer literacy,” which Brey notes is an important quality in today’s market place.

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Gender Bias in Video Games

- Brey also notes that many critics argue that the computer industry is mainly to blame for the gender gap that exists.
 - For example, critics point out that most game developers are male.
 - Also, there may be little interest on the part of software developers to design appropriate video games for women.
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Gender Bias in Video Games

- Brey notes that very few computer games include “decent role models” for women.
 - For example, he argues that “a disproportionate number of the female characters” in these games are strippers or prostitutes and that these characters tend to have “unrealistic body images.”
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Gender Bias in Video Games

- It would seem that both Brey and Buchanan make a plausible case for:
 - how the assumptions of developers designing video games can contribute to gender bias;
 - why gender bias is morally significant.
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